

Soroban Block Explorer — Milestone 1 Deliverable Evidence

- **Project:** Soroban Block Explorer
- **Team:** Rumble Fish

This document is the full written companion to the Milestone 1 submission video. It maps every acceptance criterion from §7.4 of the technical design to concrete on-mainnet evidence — resource names, SQL queries with output, screenshots, and code references. It also documents the one mid-tranche scope refinement (PostgreSQL on AWS RDS → ClickHouse on Hetzner) honestly, with cost and fit rationale grounded in ADR 0047.

Screenshot placeholders in this source are replaced with inline evidence images in the published PDF.

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1. Executive summary

Milestone 1 — **Indexing Pipeline & Core Infrastructure** — is complete and running on Stellar mainnet. The system reads every ledger from the network, parses the full XDR payload (including Soroban contract invocations and CAP-67 contract events), and writes typed rows into our own database. There is **no third-party chain API on the live read path**: the only inputs are Stellar network peers (consumed by Galexie via Captive Core) and the public Stellar history archive (consumed only by the one-time backfill).

End-to-end, the pipeline closes a ledger on mainnet and writes it to our database in well under ten seconds. The database holds a **gap-free history from the Soroban-mainnet activation ledger to the current tip**, and every CAP-67 event is stored as one decoded row with topics and data fields — not as raw XDR. The infrastructure is defined entirely as code (AWS CDK plus Ansible). Monitoring and production alarms are in place and currently healthy.

One mid-tranche scope refinement is worth disclosing up front: the production datastore was originally specified as PostgreSQL on AWS RDS and is now ClickHouse on a Hetzner dedicated server. The deliverable scope — gap-free mainnet indexing with full Soroban event data — is unchanged; details and rationale are in [section 4](#).

2. Deliverable definition

Verbatim from [docs/architecture/technical-design-general-overview.md §7.4](#) ("Three-Milestone Delivery Plan → Deliverable 1"):

Deliverable 1 — Indexing Pipeline & Core Infrastructure

Galexie ECS Fargate task running on mainnet, writing LedgerCloseMeta XDR files to S3 every ~5–6 seconds. Lambda Ledger Processor triggered per file, parsing and writing ledgers, transactions, operations, accounts, Soroban invocations, and CAP-67 events to a dedicated **ClickHouse on Hetzner** database (ch-prod-01, single-

node MergeTree, mTLS behind Caddy). Historical backfill from Soroban mainnet activation ledger (late 2023) delivered via FREEZE + rsync + ATTACH PART transport per ADR 0045. Rust API scaffolding with core modules (axum + utoipa). OpenAPI specification. AWS CDK infrastructure-as-code (AWS side) plus an Ansible playbook for the Hetzner database host. CloudWatch dashboards and ingestion lag alarms.

Acceptance criteria:

1. S3 bucket contains consecutive LedgerCloseMeta files with timestamps matching mainnet ledger close times.
2. **ClickHouse on Hetzner** ledgers table contains all ledgers from backfill start through current tip with no gaps.
3. **ClickHouse on Hetzner** soroban_events table contains full-content rows for CAP-67 events in known Soroswap / Aquarius / Phoenix transactions (spot-checked by transaction hashes); decoded events are confirmed by fetching the corresponding .xdr.zst from the public archive and re-expanding via `xdr_parser::extract_events`.
4. cdk deploy (AWS side) + ansible-playbook (Hetzner side) from clean environments produces the full working stack with no manual steps.
5. CloudWatch dashboard accessible; Galexie lag alarm fires correctly in staging.

Editorial note on the "late 2023" date in the prose above: this matches the in-tree §7.4 text we are quoting verbatim. The Soroban-protocol release (Protocol 20) actually activated on Stellar mainnet on 2024-02-20; the in-tree wording will be corrected in a follow-up documentation pass. The on-mainnet activation ledger sequence is unaffected.

Section 5 of this document walks through each acceptance criterion in turn and shows the concrete evidence that it is met.

3. Architecture

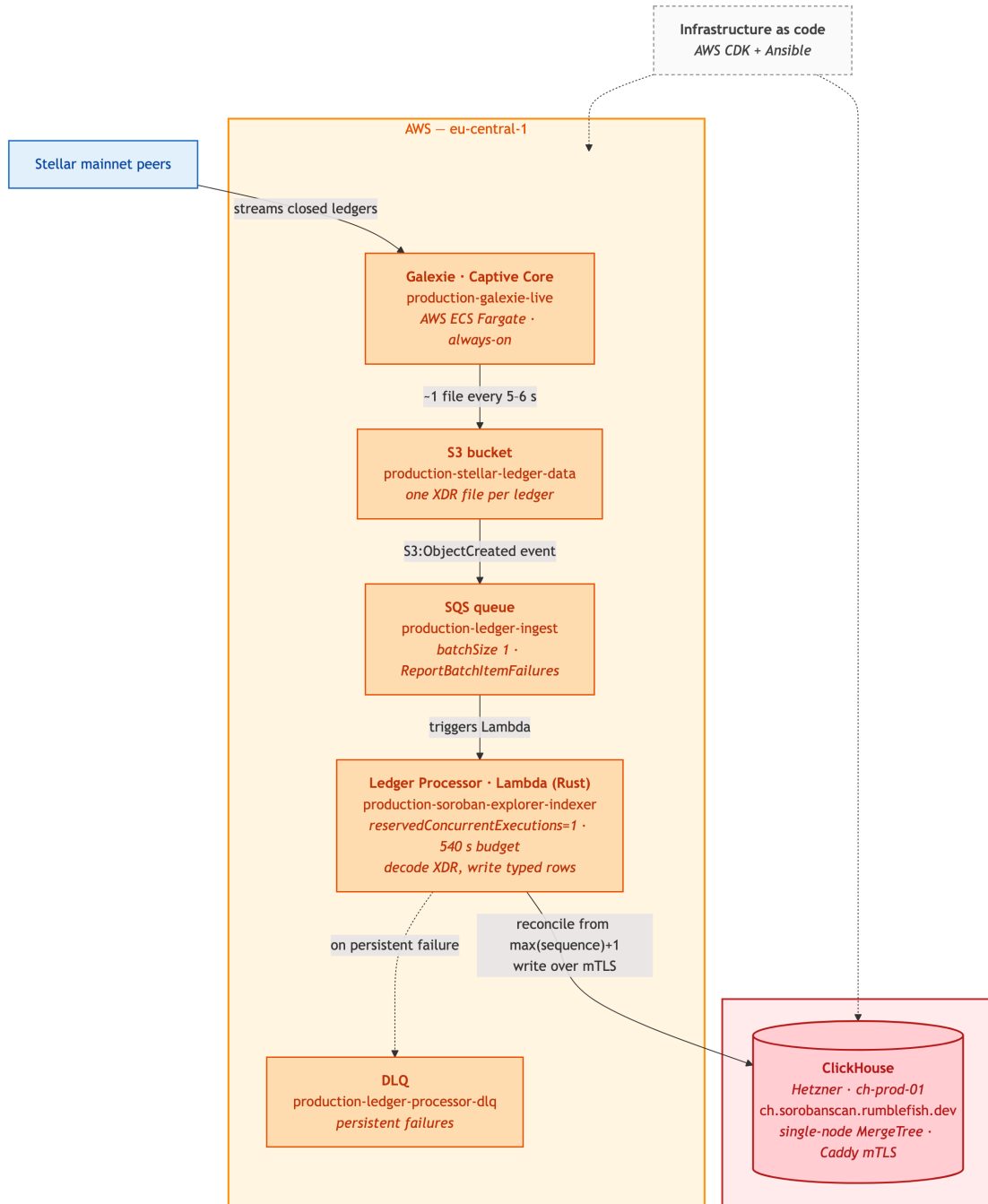


Figure 1 — Milestone 1 production indexing architecture: Galexie exports mainnet ledger files to S3, the Rust Ledger Processor reconciles and writes typed rows, and ClickHouse on Hetzner serves as the primary datastore.

Why this shape:

- **S3 between Galexie and the processor** makes the pipeline replayable. Every ledger is a durable file; a failed run does not lose a ledger because the Ledger Processor reconciles forward from $\max(\text{sequence}) + 1$ in ClickHouse on every invocation, replaying any S3 objects the previous run did not persist.
- **Galexie on Fargate**, not Lambda, because Captive Core holds an open connection to Stellar peers and resumes from a checkpoint after restart. Long-running with persistent state — a poor fit for Lambda's execution model.
- **Lambda for the processor**, driven by an **SQS doorbell + ClickHouse cursor reconcile** (task 0241): an S3 ObjectCreated notification enqueues a doorbell SQS message, one Lambda invocation persists the contiguous run from $\max(\text{sequence}) + 1$ oldest-first, and stops when the next ledger is absent on S3 (gap) or the 540 s time budget

is reached. `reservedConcurrentExecutions = 1` guarantees ascending, gapless ordering without needing an SQS FIFO queue; per-invocation state lives in ClickHouse, so the Lambda itself is stateless and freely re-runnable. End-to-end per ledger — file download, XDR decode, typed INSERTs — runs in **about half a second**. SQS handles transient retries automatically; persistent failures land in a dead-letter queue (`production-ledger-processor-dlq`), so no ledger is lost silently.

- **Hybrid AWS + Hetzner** is deliberate. Compute stays on AWS. The database lives on Hetzner over mTLS — the Lambda does not care where the database is. Cost rationale in [section 4](#).

The full architecture document is at [docs/architecture/technical-design-general-overview.md](#).

4. Scope refinement: PostgreSQL on RDS → ClickHouse on Hetzner

What changed

The approved Milestone 1 plan specified PostgreSQL 16 on AWS RDS as the primary datastore. During the milestone we migrated the primary datastore to **ClickHouse 24 on a Hetzner dedicated server (ch-prod-01)**, accessed by the Lambda over mutually-authenticated TLS behind a Caddy reverse proxy.

The decision is formally recorded in [ADR 0047 — ClickHouse on Hetzner as primary API datastore](#). Related ADRs:

- [ADR 0044](#) — initial ClickHouse pilot as parallel store
- [ADR 0045](#) — historical backfill transport (local CH → Hetzner via FREEZE + rsync + ATTACH PART)

Why we changed it

Workload fit. A block explorer is overwhelmingly read-heavy, and almost every page is a time-range scan: "recent transactions for account X", "events for contract Y in the last 24 h", "ledger N and the ten before it". ClickHouse is a columnar OLAP engine designed for exactly this access pattern. Blockchain data also compresses well in column form because the same account IDs, contract addresses, and operation types repeat heavily. Our full mainnet history occupies **~700 GB in ClickHouse** versus an estimated **~8 TB in PostgreSQL** — roughly an order of magnitude smaller.

Operating cost. At eight terabytes, the cost gap is large. An RDS instance class sized for the working set runs **over \$800 per month for the database alone**, before backups, IOPS provisioning, or read replicas. The Hetzner setup — dedicated server plus storage box for backups — totals **about \$140 per month**. The compute stays on AWS, so the saving is purely on the data tier and is sustained, not one-off.

What did not change

- The deliverable scope: gap-free mainnet indexing with full Soroban event data. Both schemas (PG and CH) carry the same logical model — ledgers, transactions, operations, account changes, Soroban invocations, CAP-67 events.
- The writer code. The historical backfill and the live Lambda share the same Rust writer implementation; the storage swap was a target change, not an algorithm change.

What this means for the form's "match approved" note

Acceptance criteria 2 and 3 in the technical design now read "ClickHouse on Hetzner" instead of "PostgreSQL on RDS" — the engine name changed, the data shape and verifiable behaviour did not. The §7.4 source has been updated in-tree (see the dated note at the top of the §7.4 block) so the documentation and the reality match.

5. Acceptance-criteria evidence

Each subsection here corresponds to one acceptance criterion from §7.4 and points to the concrete artefact that proves it. The submission video walks through these in the same order.

AC 1 — S3 ledger stream

"S3 bucket contains consecutive LedgerCloseMeta files with timestamps matching mainnet ledger close times."

Resources:

- ECS cluster: production-ingestion
- ECS service: production-galexie-live (desired = running = 1, always-on)
- CloudWatch log group: /ecs/production/galexie-live
- S3 bucket: production-stellar-ledger-data
- Region: eu-central-1

How to verify:

1. AWS Console → ECS → cluster production-ingestion → service production-galexie-live: one running task, no recent restarts.
2. CloudWatch Logs → /ecs/production/galexie-live: ledgers being exported one by one with timestamps 5–6 seconds apart (mainnet close cadence).
3. S3 → bucket production-stellar-ledger-data, sorted by "Last modified" descending: a continuous stream of <sequence>.xdr.zst objects with timestamps seconds apart. Refresh the listing — new files appear at the top in real time.

Current catch-up mode. At the time this evidence was captured, Galexie was intentionally running from the end of the imported ClickHouse backfill toward the current live network head. That means the S3 bucket can receive many ledger files in quick succession while Galexie closes the remaining gap. Once it catches up to the live head, the same always-on task settles into the normal mainnet cadence: one new LedgerCloseMeta file roughly every 5-6 seconds. At the current catch-up rate, Galexie is expected to reach the live head within approximately two days.

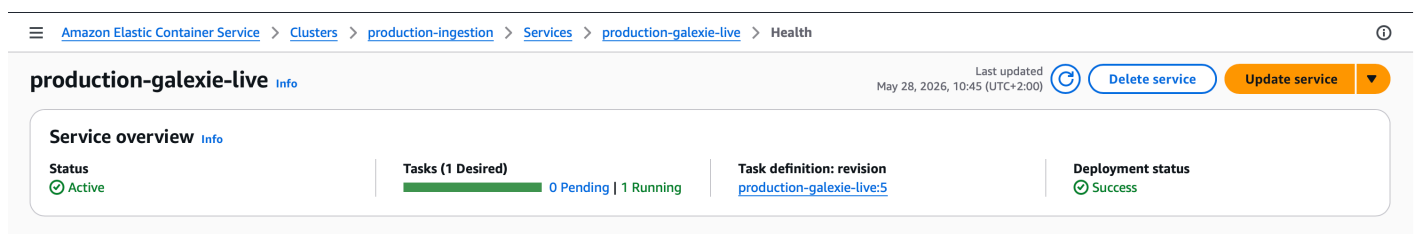


Figure 2 — ECS service production-galexie-live is active with one desired task and one running task.

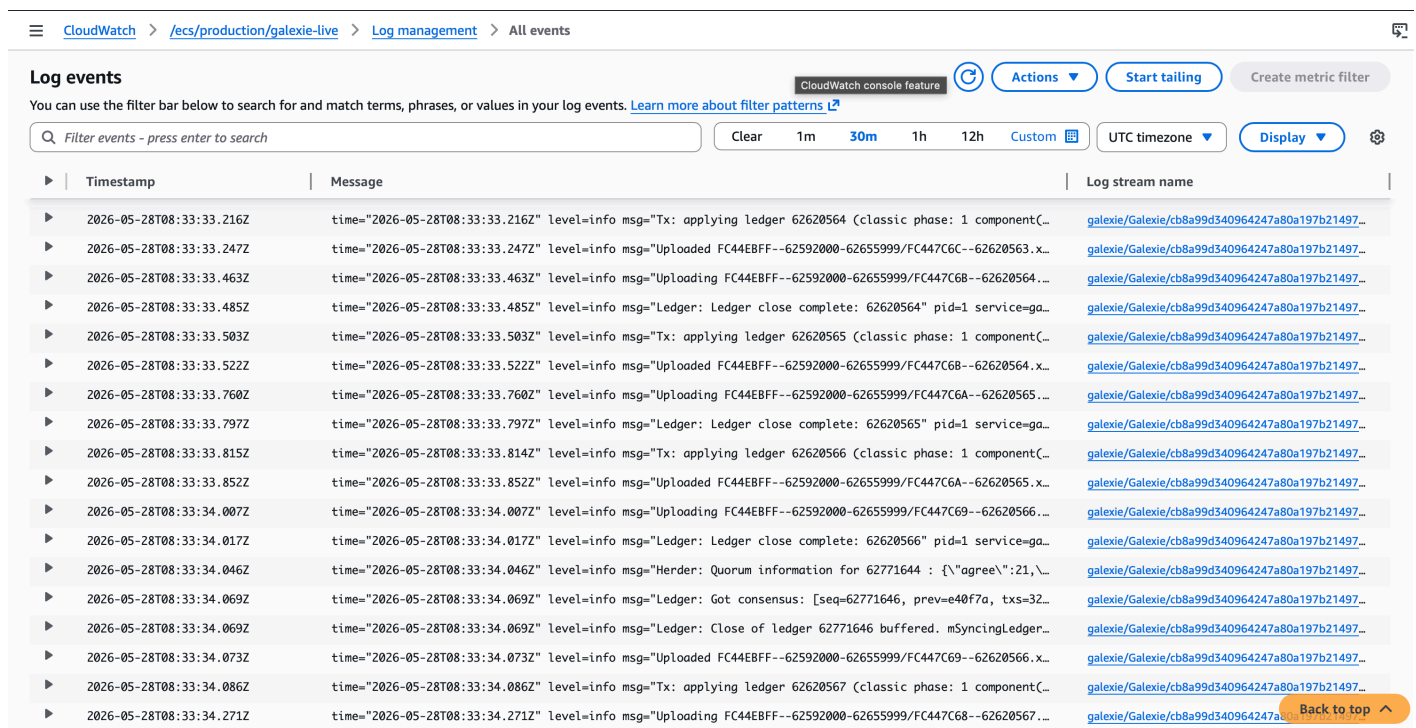


Figure 3 — CloudWatch log events show Galexie exporting consecutive mainnet ledgers into the S3-backed ingestion path.

Amazon S3 > Buckets > production-stellar-ledger-data > FC44EBFF--62592000-62655999/

FC44EBFF--62592000-62655999/ [Copy S3 URI](#)

Objects Properties

To enable sorting in the table below, use the search to reduce the size of the list to 999 objects or fewer.

Objects (999+) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	FC4473A3--62622812.xdr.zst	zst	May 28, 2026, 10:52:04 (UTC+02:00)	164.4 KB	Standard
☐	FC4473A4--62622811.xdr.zst	zst	May 28, 2026, 10:52:04 (UTC+02:00)	150.8 KB	Standard
☐	FC4473A5--62622810.xdr.zst	zst	May 28, 2026, 10:52:04 (UTC+02:00)	157.9 KB	Standard
☐	FC4473A6--62622809.xdr.zst	zst	May 28, 2026, 10:52:04 (UTC+02:00)	149.9 KB	Standard
☐	FC4473A7--62622808.xdr.zst	zst	May 28, 2026, 10:52:03 (UTC+02:00)	171.1 KB	Standard
☐	FC4473A8--62622807.xdr.zst	zst	May 28, 2026, 10:52:03 (UTC+02:00)	175.8 KB	Standard
☐	FC4473A9--62622806.xdr.zst	zst	May 28, 2026, 10:52:03 (UTC+02:00)	191.7 KB	Standard

Figure 4 — S3 bucket *production-stellar-ledger-data* contains fresh *.xdr.zst* ledger objects ordered by latest modification time.

Why S3 is on the live path. Every ledger is a durable file. The Ledger Processor is woken by an SQS doorbell and reconciles forward from $\max(\text{sequence}) + 1$ in ClickHouse on every invocation, replaying any S3 objects the previous run did not persist — no data is lost silently, and a transient failure costs at most one extra Lambda warm-up before the gap closes. SQS handles transient retries automatically; persistent failures land in a dead-letter queue (*production-ledger-processor-dlq*) for manual replay, so no ledger is dropped without an alarm trail. The full per-ledger path — download, XDR decode, typed INSERTs into ClickHouse — runs in about half a second.

AC 2 — Gap-free ClickHouse ledger history

"ClickHouse on Hetzner ledgers table contains all ledgers from backfill start through current tip with no gaps."

Endpoint: `ch.sorobanscan.rumblefish.dev` (mTLS-gated; client certificates issued to reviewers on request).

Coverage check. How many distinct ledgers we hold and the range we cover:

```
SELECT count(DISTINCT sequence) AS distinct_ledgers,
       min(sequence)           AS first_ledger,
       max(sequence)           AS tip
FROM ledgers;
```

Expected: `first_ledger` is the Soroban-mainnet activation ledger, `tip` is within seconds of network head, and `distinct_ledgers` equals $(\text{tip} - \text{first_ledger} + 1)$.

Latest captured output from `komendy.txt` (captured while live ingestion was still advancing):

distinct_ledgers	first_ledger	tip
12154127	50457424	62611550

1 row in set. Elapsed: 0.196 sec. Processed 12.15 million rows, 97.23 MB (62.08 million rows/s.,

496.63 MB/s.)

Peak memory usage: 451.27 MiB.

Output 1 — Coverage query over ledgers: 12,154,127 distinct ledgers indexed from sequence 50,457,424 through 62,611,550 at capture time.

The Milestone-1 proof — completeness. This single query is the gap-free guarantee:

```
SELECT (max(sequence) - min(sequence) + 1) AS expected_span,  
       count(DISTINCT sequence)           AS distinct_ledgers,  
       expected_span - distinct_ledgers    AS missing  
FROM ledgers;
```

Expected: missing = 0.

Latest captured output from komendy.txt:

expected_span	distinct_ledgers	missing
12154233	12154233	0

1 row in set. Elapsed: 0.203 sec. Processed 12.15 million rows, 97.23 MB (60.02 million rows/s., 480.16 MB/s.)

Peak memory usage: 451.17 MiB.

Output 2 — Completeness query over ledgers: expected span equals distinct ledger count and missing = 0.

Live tail. Run the same query repeatedly while ingestion is active — the top sequence advances:

```
SELECT sequence, closed_at, transaction_count  
FROM ledgers  
ORDER BY sequence DESC  
LIMIT 10;
```

Latest captured outputs from komendy.txt:

First capture:

sequence	closed_at	transaction_count
62611722	2026-05-17 16:18:12.000	373
62611721	2026-05-17 16:18:06.000	392
62611720	2026-05-17 16:18:00.000	378
62611719	2026-05-17 16:17:55.000	357
62611718	2026-05-17 16:17:49.000	382
62611717	2026-05-17 16:17:43.000	398
62611716	2026-05-17 16:17:37.000	393
62611715	2026-05-17 16:17:32.000	423
62611714	2026-05-17 16:17:26.000	313
62611713	2026-05-17 16:17:20.000	265

10 rows in set. Elapsed: 0.005 sec. Processed 32.78 thousand rows, 655.66 KB (5.99 million rows/s., 119.87 MB/s.)

Peak memory usage: 4.31 MiB.

Output 3 — First live-tail capture: latest returned ledger was 62,611,722.

Second capture from the same query while ingestion continued:

sequence	closed_at	transaction_count
62611740	2026-05-17 16:19:56.000	377
62611739	2026-05-17 16:19:50.000	360

62611738	2026-05-17 16:19:45.000	374
62611737	2026-05-17 16:19:39.000	369
62611736	2026-05-17 16:19:33.000	329
62611735	2026-05-17 16:19:27.000	382
62611734	2026-05-17 16:19:21.000	303
62611733	2026-05-17 16:19:16.000	228
62611732	2026-05-17 16:19:10.000	229
62611731	2026-05-17 16:19:05.000	295

10 rows in set. Elapsed: 0.006 sec. Processed 32.80 thousand rows, 656.02 KB (5.36 million rows/s., 107.21 MB/s.)

Peak memory usage: 4.30 MiB.

Output 4 — Second live-tail capture: latest returned ledger advanced to 62,611,740 while ingestion continued.

How the historical range was populated. Running the live pipeline back through every ledger since Soroban activation would saturate the cloud path with one-at-a-time S3-event invocations for millions of historical ledgers. Instead, we indexed history on local machines (using the same Rust writer code as the live Lambda) and transported the resulting ClickHouse parts onto the Hetzner server with the FREEZE + rsync + ATTACH PART procedure documented in [ADR 0045](#). This is purely a transport optimisation — the parsing logic is identical.

AC 3 — Full-content CAP-67 Soroban events

"ClickHouse on Hetzner soroban_events table contains full-content rows for CAP-67 events in known Soroswap / Aquarius / Phoenix transactions (spot-checked by transaction hashes); decoded events are confirmed by fetching the corresponding .xdr.zst from the public archive and re-expanding via xdr_parser::extract_events."

Schema (relevant columns of soroban_events, defined in [crates/db-clickhouse/schema/init.sql](#)). Despite the _xdr suffix on topics_xdr and data_xdr, these columns hold **decoded JSON** (ScVal-decoded by the indexer, ZSTD-compressed for storage). The suffix is historical from an earlier raw-XDR draft of the schema; the codec comment in init.sql documents the actual content.

Column	Meaning
ledger_sequence	Ledger the event belongs to
transaction_id	FK to transactions (surrogate key)
contract_id	FK to soroban_contracts (surrogate key)
event_index	Ordinal of the event within its transaction
event_type	contract / system / diagnostic
signature	Event signature (function-name-like topic 0 for typed events)
topics_xdr	Decoded topics (one row per event, not raw XDR concat)
data_xdr	Decoded data payload

Table 1 — Relevant soroban_events columns used to verify full decoded CAP-67 event content.

Spot-check by contract. The contract picked for this spot-check is CBQDHNBFZYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK — the Aquarius soroswap-style multi-pool router on mainnet, one of the contracts the AC wording calls out by name.


```

SELECT e.ledger_sequence,
       e.event_index,
       e.event_type,
       e.signature,
       e.topics_xdr,
       e.data_xdr
FROM   soroban_events AS e
WHERE  e.contract_id = (
        SELECT id
        FROM   soroban_contracts FINAL
        WHERE  contract_id = 'CBQDHNBFBZYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK'
        LIMIT  1
      )
ORDER  BY e.ledger_sequence DESC
LIMIT  20
SETTINGS optimize_read_in_order = 1
FORMAT Vertical;

```

The StrKey → surrogate-Int64 resolution is moved into a scalar subquery against soroban_contracts FINAL (PK lookup by contract_id StrKey, microseconds). Substituting the resolved constant lets the main statement use the soroban_events primary-key skip index (contract_id, ledger_sequence, transaction_id, event_index) and optimize_read_in_order = 1 to stop the reverse scan as soon as the first 20 rows are produced — no global sort. An INNER JOIN soroban_contracts form would block that pushdown because ClickHouse does not propagate values from a join's right-hand side into the left-hand side's PK skip index.

Result — 20 most-recent CAP-67 events for the Aquarius router (all of them swaps on different pool counterparties; the signature, topics_xdr and data_xdr columns are the ScVal-decoded JSON the indexer wrote, not raw XDR):

Row 1:

```

ledger_sequence: 62611855 -- 62.61 million
event_index:     17
event_type:      1
signature:       swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CC5UXAGZOU270QBKBYTQMES3NV06EV6FCMWSNPPHAPIS6S24ENM3C24A"
      }
    ]
  },
  {
    "type": "address",
    "value": "GB7JRBCGTLHSWSIBLTXMJJDQF700YEEEEBXJ0BWIA2RJKG3GGCWG32X"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CAJXB0GWSRFT7Q7ZKHVTPWG0DOBBSPTYQVKN2WSMN2WFMPAXX2CETEBAZ"
    },
    {
      "type": "address",
      "value": "CC5UXAGZ0U270QBKYTQMES3NV06EV6FCMWSNPPHAPIS6S24ENM3C24A"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "2854455340306"
    },
    {
      "type": "u128",
      "value": "40507804"
    }
  ]
}

```

Row 2:

```

ledger_sequence: 62611855 -- 62.61 million
event_index:      12
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CBLLEW7HD2RWATVSMLAGWM4G3WCHSHDJ25ALP4DI6LULV5TU35N2CIZA"
      },
      {
        "type": "address",
        "value": "CC5UXAGZ0U270QBKYTQMES3NV06EV6FCMWSNPPHAPIS6S24ENM3C24A"
      }
    ]
  },
  {
    "type": "address",
    "value": "GB7JRBCGTLHSWSIBLTXOMXJDQF700YEEEEBXJ0BWIA2RJKG3GGCWG32X"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CD2UBY4D0GL3FETXUNMSSEGGKQAL35ABFG6WM04P24GK2SH3GULFRCKFA"
    },
    {
      "type": "address",
      "value": "CBLLEW7HD2RWATVSMLAGWM4G3WCHSHDJ25ALP4DI6LULV5TU35N2CIZA"
    },
    {
      "type": "address",
      "value": "CC5UXAGZ0U270QBKBYTQMES3NV06EV6FCMWSNPPHAPIS6S24ENM3C24A"
    },
    {
      "type": "u128",
      "value": "249567626"
    },
    {
      "type": "u128",
      "value": "2854455340306"
    }
  ]
}

```

Row 3:

```

ledger_sequence: 62611855 -- 62.61 million
event_index:      7
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
      },
      {
        "type": "address",
        "value": "CBLLEW7HD2RWATVSMLAGWM4G3WCHSHDJ25ALP4DI6LULV5TU35N2CIZA"
      }
    ]
  },
  {
    "type": "address",
    "value": "GB7JRBCGTLHSWSIBLTX0MXJDQF700YEEEEBXJ0BWIA2RJKG3GGCWG32X"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBFXFTFLUUIBQGHLMSI762JUD34NHLFZ5AGPLEQ2WJ2CEQANDPZ0DNVN"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CBLLEW7HD2RWATVSMLAGWM4G3WCHSHDJ25ALP4DI6LULV5TU35N2CIZA"
    },
    {
      "type": "u128",
      "value": "18832335666"
    },
    {
      "type": "u128",
      "value": "249567626"
    }
  ]
}

```

Row 4:

```

ledger_sequence: 62611851 -- 62.61 million
event_index:      6
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
      },
      {
        "type": "address",
        "value": "CBMFDIRY50KI4JJURXC4SMEQPWB4UUADIADJK4NA6CYBN0YK4W4TMLLF"
      }
    ]
  },
  {
    "type": "address",
    "value": "GDERSSCKJQPPXUQ0ZIOXGRVAGNLVPVZCJ2MAX7RCMVMWGRPVAEG7XGTK"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CAMXZXXBD7DFBLYLHUW24U4MY37X7SU5XXT5ZVVUBXRXWLAIM7INI7G2"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CBMFDIRY50KI4JJURXC4SMEQPWB4UUADIADJK4NA6CYBNOYK4W4TMLLF"
    },
    {
      "type": "u128",
      "value": "2093920028"
    },
    {
      "type": "u128",
      "value": "2094304941"
    }
  ]
}

```

Row 5:

```

ledger_sequence: 62611847 -- 62.61 million
event_index:      6
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
      },
      {
        "type": "address",
        "value": "CBMFDIRY50KI4JJURXC4SMEQPWB4UUADIADJK4NA6CYBNOYK4W4TMLLF"
      }
    ]
  },
  {
    "type": "address",
    "value": "GDERSSCKJQPPXUQ0ZI0XGRVAGNLVPVZCJ2MAX7RCMVMWGRPVAEG7XGTK"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CAMXZXXBD7DFBLYLHUW24U4MY37X7SU5XXT5ZVVUBXRXWLAIM7INI7G2"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CBMFDIRY50KI4JJURXC4SMEQPWB4UUADIADJK4NA6CYBNOYK4W4TMLLF"
    },
    {
      "type": "u128",
      "value": "8750416245"
    },
    {
      "type": "u128",
      "value": "8752025410"
    }
  ]
}

```

Row 6:

```

ledger_sequence: 62611846 -- 62.61 million
event_index:      15
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBBMQBNHB2FYVZYV7VNH0JHUMTFJLR4PUMRVQYNW6RHIKZ02NQMIBUCV"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "1502939385"
    },
    {
      "type": "u128",
      "value": "9931549831"
    }
  ]
}

```

Row 7:

```

ledger_sequence: 62611846 -- 62.61 million
event_index:      9
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```



```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBRUQ7I6C60GHMDYWD6XQUZFB6KJ3LLPNE34EPKSPFZ2YMBJ2GIWYYZ7"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "u128",
      "value": "4641523835859"
    },
    {
      "type": "u128",
      "value": "1502939385"
    }
  ]
}

```

Row 8:

```

ledger_sequence: 62611834 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CDE57N6XTUPBKYYDGMXX7E7SLN0LFY3JEB4MULSMR2AKTSAENGX2HC"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "77976866491"
    },
    {
      "type": "u128",
      "value": "167051949"
    }
  ]
}

```

Row 9:

```

ledger_sequence: 62611820 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCKCKCPHYVXD4NECBFJTFSCU2AMSJGCNG406K4JVRE2BLPR7WNDBQIQ"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CD65ER0VLTDU2DWM4ZUJF4NHK4A46DX2UA0GCV7YDFPCSLFYNH57KGIY"
    },
    {
      "type": "address",
      "value": "CCKCKCPHYVXQD4NECBFJTFSCU2AMSJGCNG406K4JVRE2BLPR7WNDBQIQ"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "10000000000000"
    },
    {
      "type": "u128",
      "value": "29799312831"
    }
  ]
}

```

Row 10:

```

ledger_sequence: 62611813 -- 62.61 million
event_index:      9
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBBMQBNHB2FYVZYV7VNH0JHUMTFJLR4PUMRVQYNW6RHIKZ02NQMIBUCV"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "300185000"
    },
    {
      "type": "u128",
      "value": "1984363241"
    }
  ]
}

```

Row 11:

```

ledger_sequence: 62611808 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCKCKCPHYVXQD4NECBFJTFSCU2AMSJGCNG406K4JVRE2BLPR7WNDBQIQ"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CD65ER0VLTDU2DWM4ZUJF4NHK4A46DX2UA0GCV7YDFPCSLFYNH57KGIY"
    },
    {
      "type": "address",
      "value": "CCKCKCPHYVXQD4NECBFJTFSCU2AMSJGCNG406K4JVRE2BLPR7WNDBQIQ"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "10000000000000"
    },
    {
      "type": "u128",
      "value": "29955756534"
    }
  ]
}

```

Row 12:

```

ledger_sequence: 62611799 -- 62.61 million
event_index:      14
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBBMQBNHB2FYVZYV7VNH0JHUMTFJLR4PUMRVQYNW6RHIKZ02NQMIBUCV"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "387778951"
    },
    {
      "type": "u128",
      "value": "2563753038"
    }
  ]
}

```

Row 13:

```

ledger_sequence: 62611799 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CA6GAF0JCW4MGQQBUCQUSA3CLIH25G4SNKB2JHYKZCVWZTNW5VXMSC40"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "u128",
      "value": "1197455089607"
    },
    {
      "type": "u128",
      "value": "387778951"
    }
  ]
}

```

Row 14:

```

ledger_sequence: 62611794 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CDE57N6XTUPBKYYDGMXX7E7SLN0LFY3JEB4MULSMR2AKTSAENGX2HC"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "63174729433"
    },
    {
      "type": "u128",
      "value": "135377697"
    }
  ]
}

```

Row 15:

```

ledger_sequence: 62611788 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "GBXD6YV6ZLMSQZDTRAAIWSG53CKNPXU2INSX4AZRFASAE3ETEWDSHULZ"
  }
]

```



```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBBMQBNHB2FYVZYV7VNH0JHUMTFJLR4PUMRVQYNW6RHIKZ02NQMI BUCV"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "u128",
      "value": "6169670000"
    },
    {
      "type": "u128",
      "value": "931425803"
    }
  ]
}

```

Row 16:

```

ledger_sequence: 62611785 -- 62.61 million
event_index:      13
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CA6PUJLBYKZKUEKLZJMKBZLEKP20THANDE0WSFF44FTSYLKQPIICCJBE"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "9587380725"
    },
    {
      "type": "u128",
      "value": "63356968062"
    }
  ]
}

```

Row 17:

```

ledger_sequence: 62611785 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      },
      {
        "type": "address",
        "value": "CD0FW7HNKLUZRLFZST4EW7V3AV4JI5IHMT6BPXXSY2IEFZ4NE5TWU2P4"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CCWNKTTMHZ4VRR3FYQT3G253MH0GRLUEVT06YPENM7ZN65DGC345LHI6"
    },
    {
      "type": "address",
      "value": "CD0FW7HNKLUZRLFZST4EW7V3AV4JI5IHMT6BPXXSY2IEFZ4NE5TWU2P4"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "u128",
      "value": "9592891971"
    },
    {
      "type": "u128",
      "value": "9587380725"
    }
  ]
}

```

Row 18:

```

ledger_sequence: 62611755 -- 62.61 million
event_index:      14
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      },
      {
        "type": "address",
        "value": "CDFZUVS5YNLXU7VENKOUDE0HCJGKQNVUBWD7KMN6E7ZR0KPYPFLRUJFG"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBV6LZAB0AJRXHLIEKILTWBPC2AXLYVR4CGGPKZQ3PWJ5QFTPIWSWGWQ"
    },
    {
      "type": "address",
      "value": "CDFZUVS5YNLXU7VENKOUDE0HCJGKQNVUBWD7KMN6E7ZR0KPYPFLRUJFG"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "u128",
      "value": "490388422"
    },
    {
      "type": "u128",
      "value": "487349327"
    }
  ]
}

```

Row 19:

```

ledger_sequence: 62611755 -- 62.61 million
event_index:      8
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAESLMGW5LYTIEJI7FJHK6SFSWRELLNVX5Q4WR4UZEALMTRWQDBKDPAG"
      },
      {
        "type": "address",
        "value": "CDFZUVS5YNLXU7VENKOUDE0HCJGKQNVUBWD7KMN6E7ZR0KPYPFLRUJFG"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CBHA22UACTWKPR7CPNFXYVX57QXQV0EPTWKBSWFQAPN7HB5GEBKRJB63"
    },
    {
      "type": "address",
      "value": "CAESLMGW5LYTIEJI7FJHK6SFSWRELLNVX5Q4WR4UZEALMTRWQDBKDPAG"
    },
    {
      "type": "address",
      "value": "CDFZUVS5YNLXU7VENKOUDE0HCJGKQNVUBWD7KMN6E7ZR0KPYPFLRUJFG"
    },
    {
      "type": "u128",
      "value": "131391318644"
    },
    {
      "type": "u128",
      "value": "490388422"
    }
  ]
}

```

Row 20:

```

ledger_sequence: 62611680 -- 62.61 million
event_index:      13
event_type:       1
signature:        swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CBLLEW7HD2RWATVSMLAGWM4G3WCHSHDJ25ALP4DI6LULV5TU35N2CIZA"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]

```

```

]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CB7FKGSTHP750RTIZGGMVUTQLEMVTSE0I4Q0RQPCABJSGTAATDFCE2YV"
    },
    {
      "type": "address",
      "value": "CBLLEW7HD2RWATVSMLAGWM4G3WCHSHDJ25ALP4DI6LULV5TU35N2CIZA"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "u128",
      "value": "642358535"
    },
    {
      "type": "u128",
      "value": "15704591"
    }
  ]
}

```

20 rows in set. Elapsed: 0.024 sec. Processed 264.59 thousand rows, 11.93 MB (11.22 million rows/s., 506.04 MB/s.)

Peak memory usage: 12.16 MiB.

Output 5 — Contract-level CAP-67 spot-check: 20 most recent decoded events for the Aquarius router contract, including pretty-printed decoded topics_xdr and data_xdr JSON.

Spot-check by transaction hash (matches the AC wording). Transaction 6cad2d49962ae5962722f1f90d4fd11f9e04bd644ad4873752ae1416fddd4740 is a Soroswap multi-hop trade that routed funds through the Aquarius router (CBQDHNBF...6QUK) — exactly the kind of "Soroswap / Aquarius / Phoenix" interaction the AC requires.

```

WITH tx AS (
  SELECT ledger_sequence
  FROM transaction_hash_index FINAL
  WHERE hash = unhex('6cad2d49962ae5962722f1f90d4fd11f9e04bd644ad4873752ae1416fddd4740')
  LIMIT 1
)
SELECT e.event_index,
       e.event_type,
       e.signature,
       e.topics_xdr,
       e.data_xdr
FROM soroban_events AS e
INNER JOIN transactions AS t
  ON t.id = e.transaction_id
  AND t.ledger_sequence = e.ledger_sequence
WHERE e.ledger_sequence = (SELECT ledger_sequence FROM tx)
  AND t.hash =
  unhex('6cad2d49962ae5962722f1f90d4fd11f9e04bd644ad4873752ae1416fddd4740')

```

```
ORDER BY e.event_index
FORMAT Vertical;
```

The query first resolves the transaction's ledger_sequence via the dedicated transaction_hash_index PK lookup (ORDER BY hash), then uses that value to (a) partition-prune soroban_events (PARTITION BY intDiv(ledger_sequence, 500000)) down to a single 500k-ledger part, and (b) activate the bloom-filter index idx_tx_hash_bloom on transactions.hash via a raw FixedString(32) comparison (a lower(hex(t.hash)) form would defeat the bloom filter because a function call on the indexed column blocks index use).

Result — all 27 CAP-67 events emitted by that transaction, in their natural event_index order. The trade chains fee → transfer → trade → update_reserves → swap four times across the pools yUSDC → AQUA → PYUSD → USDC → native, finishing with a charge_provider_fee:

Row 1:

```
event_index: 0
event_type: 1
signature: fee
topics_xdr:
[
  {
    "type": "sym",
    "value": "fee"
  },
  {
    "type": "address",
    "value": "GBUX7QKLFWWDJZH5MZEENVFZNP6ZXYYF5ILG2IPFY2LJEAUW20PHDQY"
  }
]
data_xdr:
{
  "type": "i128",
  "value": "171241"
}
```

Row 2:

```
event_index: 1
event_type: 1
signature: fee
topics_xdr:
[
  {
    "type": "sym",
    "value": "fee"
  },
  {
    "type": "address",
    "value": "GBUX7QKLFWWDJZH5MZEENVFZNP6ZXYYF5ILG2IPFY2LJEAUW20PHDQY"
  }
]
data_xdr:
{
  "type": "i128",
  "value": "-14854"
}
```

Row 3:

```
event_index: 2
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "GBUX7QKLFWDJZH5MZEENVFZNP6ZXYYF5ILG2IPFY2LJEAUW20PHDQY"
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  },
  {
    "type": "string",
    "value": "yUSDC:GDGTWSM4MGS4T7Z6W4RPWOCHE2I6RDFCIFYZGS3D0A63LWQTRNZNTTFF"
  }
]
data_xdr:
{
  "type": "i128",
  "value": "18175"
}
```

Row 4:

```
event_index: 3
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "string",
    "value": "yUSDC:GDGTWSM4MGS4T7Z6W4RPWOCHE2I6RDFCIFYZGS3D0A63LWQTRNZNTTFF"
  }
]
data_xdr:
{
  "type": "i128",
```



```
    "value": "18175"
  }
```

Row 5:

```
event_index: 4
event_type: 1
signature: transfer
topics_xdr:
  [
    {
      "type": "sym",
      "value": "transfer"
    },
    {
      "type": "address",
      "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
    },
    {
      "type": "address",
      "value": "CCBB55HJJQN0A56FT37SI3B6EYJ6VMW2RB4BZJZGTEZBT5QUMN3DUGCQ"
    },
    {
      "type": "string",
      "value": "yUSDC:GDGTVWSM4MGS4T7Z6W4RPWOCHE2I6RDFCIFYZGS3D0A63LWQTRNZNTTFF"
    }
  ]
data_xdr:
  {
    "type": "i128",
    "value": "18175"
  }
```

Row 6:

```
event_index: 5
event_type: 1
signature: transfer
topics_xdr:
  [
    {
      "type": "sym",
      "value": "transfer"
    },
    {
      "type": "address",
      "value": "CCBB55HJJQN0A56FT37SI3B6EYJ6VMW2RB4BZJZGTEZBT5QUMN3DUGCQ"
    },
    {
      "type": "address",
      "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
    },
    {
      "type": "string",
      "value": "AQUA:GBNZILSTVQZ4R7IKQDGHYGY2QXL5Q0FJYQMPKWRRM5PAV7Y4M67AQUA"
    }
  ]
```

```
data_xdr:
{
  "type": "i128",
  "value": "51328912"
}
```

Row 7:

```
event_index: 6
event_type: 1
signature: trade
topics_xdr:
[
  {
    "type": "sym",
    "value": "trade"
  },
  {
    "type": "address",
    "value": "CD0FW7HNKLUFZST4EW7V3AV4JI5IHMT6BPXXSY2IEFZ4NE5TWU2P4"
  },
  {
    "type": "address",
    "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  }
]
```

```
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "i128",
      "value": "18175"
    },
    {
      "type": "i128",
      "value": "51328912"
    },
    {
      "type": "i128",
      "value": "10"
    }
  ]
}
```

Row 8:

```
event_index: 7
event_type: 1
signature: update_reserves
topics_xdr:
[
  {
```

```

        "type": "sym",
        "value": "update_reserves"
    }
]
data_xdr:
{
    "type": "vec",
    "value": [
        {
            "type": "i128",
            "value": "69962120443494"
        },
        {
            "type": "i128",
            "value": "24746933502"
        }
    ]
}

```

Row 9:

```

event_index: 8
event_type: 1
signature: swap
topics_xdr:
[
    {
        "type": "sym",
        "value": "swap"
    },
    {
        "type": "vec",
        "value": [
            {
                "type": "address",
                "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
            },
            {
                "type": "address",
                "value": "CD0FW7HNKLUFZST4EW7V3AV4JI5IHMT6BPXXSY2IEFZ4NE5TWU2P4"
            }
        ]
    },
    {
        "type": "address",
        "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
    }
]
data_xdr:
{
    "type": "vec",
    "value": [
        {
            "type": "address",
            "value": "CCBB55HJJQN0A56FT37SI3B6EYJ6VMW2RB4BZJZGTEZBT5QUMN3DUGCQ"
        },
        {

```

```

    "type": "address",
    "value": "CD0FW7HNKLUZRLFZST4EW7V3AV4JI5IHMT6BPXXSY2IEFZ4NE5TWU2P4"
  },
  {
    "type": "address",
    "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
  },
  {
    "type": "u128",
    "value": "18175"
  },
  {
    "type": "u128",
    "value": "51328912"
  }
]
}

```

Row 10:

```

event_index: 9
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CBQDHNBFZYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "address",
    "value": "CADFWSBBD6VMCL45DEPZ37X3JNX0ZXIWEVJJTHMQH3UEB3JSQVJSPG2I"
  },
  {
    "type": "string",
    "value": "AQUA:GBNZILSTVQZ4R7IKQDGHYGY2QXL5Q0FJYQMPKWRRM5PAV7Y4M67AQUA"
  }
]
data_xdr:
{
  "type": "i128",
  "value": "51328912"
}

```

Row 11:

```

event_index: 10
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",

```

```

    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CADFWSBBD6VMCL45DEPZ37X3JNX0ZXIWEVJJTHMQH3UEB3JSQVJSPG2I"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "string",
    "value": "PYUSD:GDQE7IXJ4HUHV6RQHIUPRJSEZE4DRS5WY57702FY6YQ5LVWZ7JZTU2V5"
  }
]
data_xdr:
{
  "type": "i128",
  "value": "18310"
}

```

Row 12:

```

event_index: 11
event_type: 1
signature: trade
topics_xdr:
[
  {
    "type": "sym",
    "value": "trade"
  },
  {
    "type": "address",
    "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
  },
  {
    "type": "address",
    "value": "CCCRWH6Q3FNP3I2I57BDLM5AFAT7060F6GKQ0C6SSJNDAVRZ57SPHGU2"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  }
]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "i128",
      "value": "51328912"
    },
    {
      "type": "i128",
      "value": "18310"
    }
  ],
}

```

```

    {
      "type": "i128",
      "value": "25665"
    }
  ]
}

```

Row 13:

```

event_index: 12
event_type: 1
signature:  update_reserves
topics_xdr:
  [
    {
      "type": "sym",
      "value": "update_reserves"
    }
  ]
data_xdr:
  {
    "type": "vec",
    "value": [
      {
        "type": "i128",
        "value": "2540357503965"
      },
      {
        "type": "i128",
        "value": "907125471"
      }
    ]
  }
}

```

Row 14:

```

event_index: 13
event_type: 1
signature:  swap
topics_xdr:
  [
    {
      "type": "sym",
      "value": "swap"
    },
    {
      "type": "vec",
      "value": [
        {
          "type": "address",
          "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
        },
        {
          "type": "address",
          "value": "CCCRWH6Q3FNP3I2I57BDLM5AFAT7060F6GKQ0C6SSJNDAVRZ57SPHGU2"
        }
      ]
    }
  ]
}

```

```

    },
    {
      "type": "address",
      "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
    }
  ]
}
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CADFWSBBD6VMCL45DEPZ37X3JNX0ZXIWEVJJTHMQH3UEB3JSQVJSPG2I"
    },
    {
      "type": "address",
      "value": "CAUIKL3IYGMERDRUN6YSCLWVAKIFG5Q4YJHUKM4S4NJZQIA3BAS60JPK"
    },
    {
      "type": "address",
      "value": "CCCRWH6Q3FNP3I2I57BDLM5AFAT7060F6GKQ0C6SSJNDAVRZ57SPHGU2"
    },
    {
      "type": "u128",
      "value": "51328912"
    },
    {
      "type": "u128",
      "value": "18310"
    }
  ]
}

```

Row 15:

```

event_index: 14
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "address",
    "value": "CDMH535JSD224YXPET3B4SJ0LXTQQ24GRSCWACGYBKSH2DKFJYWI7SUW"
  },
  {
    "type": "string",
    "value": "PYUSD:GDQE7IXJ4HUHV6RQHIUPRJSEZE4DRS5WY57702FY6YQ5LVWZ7JZTU2V5"
  }
]

```

```
data_xdr:
{
  "type": "il28",
  "value": "18310"
}
```

Row 16:

```
event_index: 15
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CDMH535JSD224YXPET3B4SJOLXTQQ24GRSCWACGYBKSH2DKFJYWI7SUW"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYWLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "string",
    "value": "USDC:GA5ZSEJYB37JRC5AVCIA5M0P4RHTM335X2KGX3IH0JAPP5RE34K4KZVN"
  }
]
data_xdr:
{
  "type": "il28",
  "value": "18290"
}
```

Row 17:

```
event_index: 16
event_type: 1
signature: trade
topics_xdr:
[
  {
    "type": "sym",
    "value": "trade"
  },
  {
    "type": "address",
    "value": "CCCRWH6Q3FNP3I2I57BDLM5AFAT7060F6GKQ0C6SSJNDAVRZ57SPHGU2"
  },
  {
    "type": "address",
    "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
  },
  {
    "type": "address",

```



```

        "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
    }
]
data_xdr:
{
    "type": "vec",
    "value": [
        {
            "type": "i128",
            "value": "18310"
        },
        {
            "type": "i128",
            "value": "18290"
        },
        {
            "type": "i128",
            "value": "19"
        }
    ]
}

```

Row 18:

```

event_index: 17
event_type: 1
signature: update_reserves
topics_xdr:
[
    {
        "type": "sym",
        "value": "update_reserves"
    }
]
data_xdr:
{
    "type": "vec",
    "value": [
        {
            "type": "i128",
            "value": "41287345834217"
        },
        {
            "type": "i128",
            "value": "39003786382504"
        }
    ]
}

```

Row 19:

```

event_index: 18
event_type: 1
signature: swap
topics_xdr:
[
    {

```

```

    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CCCRWH6Q3FNP3I2I57BDLM5AFAT7060F6GKQ0C6SSJNDAVRZ57SPHGU2"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CDMH535JSD224YXPET3B4SJ0LXTQQ24GRSCWACGYBKSH2DKFJYWI7SUW"
    },
    {
      "type": "address",
      "value": "CCCRWH6Q3FNP3I2I57BDLM5AFAT7060F6GKQ0C6SSJNDAVRZ57SPHGU2"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "u128",
      "value": "18310"
    },
    {
      "type": "u128",
      "value": "18290"
    }
  ]
}

```

Row 20:

```

event_index: 19
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",

```

```

    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "address",
    "value": "CA6PUJLBYKZKUEKLZJMKBZLEKP20THANDE0WSFF44FTSYLKQPIICCJBE"
  },
  {
    "type": "string",
    "value": "USDC:GA5ZSEJYB37JRC5AVCIA5M0P4RHTM335X2KGX3IH0JAPP5RE34K4KZVN"
  }
]
data_xdr:
{
  "type": "il28",
  "value": "18290"
}

```

Row 21:

```

event_index: 20
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CA6PUJLBYKZKUEKLZJMKBZLEKP20THANDE0WSFF44FTSYLKQPIICCJBE"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "string",
    "value": "native"
  }
]
data_xdr:
{
  "type": "il28",
  "value": "109465"
}

```

Row 22:

```

event_index: 21
event_type: 1
signature: trade
topics_xdr:

```

```
[
  {
    "type": "sym",
    "value": "trade"
  },
  {
    "type": "address",
    "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
  },
  {
    "type": "address",
    "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
  },
  {
    "type": "address",
    "value": "CBQDHNBFBZYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  }
]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "i128",
      "value": "18290"
    },
    {
      "type": "i128",
      "value": "109465"
    },
    {
      "type": "i128",
      "value": "10"
    }
  ]
}
```

Row 23:

```
event_index: 22
event_type: 1
signature: update_reserves
topics_xdr:
[
  {
    "type": "sym",
    "value": "update_reserves"
  }
]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "i128",
      "value": "135230799353474"
    }
  ],
```

```

    {
      "type": "i128",
      "value": "22571512200220"
    }
  ]
}

```

Row 24:

```

event_index: 23
event_type: 1
signature: swap
topics_xdr:
[
  {
    "type": "sym",
    "value": "swap"
  },
  {
    "type": "vec",
    "value": [
      {
        "type": "address",
        "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
      },
      {
        "type": "address",
        "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
      }
    ]
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQODRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  }
]
data_xdr:
{
  "type": "vec",
  "value": [
    {
      "type": "address",
      "value": "CA6PUJLBYKZKUEKLZJMKBZLEKP20THANDE0WSFF44FTSYLKQPIICCJBE"
    },
    {
      "type": "address",
      "value": "CCW67TSZV3SSS2HXMBQ5JFGCKJNXKZM7UQUWUZPUTHXSTZLE07SJMI75"
    },
    {
      "type": "address",
      "value": "CAS3J7GYLGXMF6TDJBBYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
    },
    {
      "type": "u128",
      "value": "18290"
    }
  ],
  {

```

```

        "type": "u128",
        "value": "109465"
    }
]
}

```

Row 25:

```

event_index: 24
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CBQDHNBFZYE4MKPWBSJ0PIYLW4SFSXAXUTSXJN76GNKYVYPCKWC6QUK"
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  },
  {
    "type": "string",
    "value": "native"
  }
]
data_xdr:
{
  "type": "i128",
  "value": "109465"
}

```

Row 26:

```

event_index: 25
event_type: 1
signature: transfer
topics_xdr:
[
  {
    "type": "sym",
    "value": "transfer"
  },
  {
    "type": "address",
    "value": "CCPGFQUTSEHDIQ0DRE3GJDNE64A35HZ32L7LPDN7GX0CIYNBJSMS6V6B"
  },
  {
    "type": "address",
    "value": "GBUX7QKLFWWDJZH5MZEENVFZNP6ZXYYF5ILG2IPFY2LJEAUW20PHDQY"
  },
  {
    "type": "string",

```

```

        "value": "native"
    }
]
data_xdr:
{
    "type": "i128",
    "value": "108917"
}

```

Row 27:

```

event_index: 26
event_type: 1
signature: charge_provider_fee
topics_xdr:
[
    {
        "type": "sym",
        "value": "charge_provider_fee"
    }
]
data_xdr:
{
    "type": "vec",
    "value": [
        {
            "type": "address",
            "value": "CAS3J7GYLGXMF6TDJBBYYSE3HQ6BBSMLNUQ34T6TZMYMW2EVH34XOWMA"
        },
        {
            "type": "u128",
            "value": "548"
        }
    ]
}

```

27 rows in set. Elapsed: 0.088 sec. Processed 4.01 million rows, 196.14 MB (45.85 million rows/s., 2.24 GB/s.)

Peak memory usage: 97.71 MiB.

Output 6 — Transaction-level CAP-67 spot-check: all 27 decoded events emitted by the selected Soroswap / Aquarius multi-hop transaction.

Independent verification. The submission can be independently re-derived from the public Stellar archive: download the `.xdr.zst` for the same ledger, feed it to `xdr_parser::extract_events` (our in-repo [crates/xdr-parser/](#), built on top of Stellar's official `stellar-xdr` Rust crate), and the resulting decoded events match what our database returns row-for-row.

AC 4 — Reproducible infrastructure as code

"cdk deploy (AWS side) + ansible-playbook (Hetzner side) from clean environments produces the full working stack with no manual steps."

Two halves, both reproducible:

Half	Definition	How it deploys
AWS	infra/ — TypeScript CDK app	cdk deploy --app "node dist/bin/production.js" --all from an operator's machine against a clean account
Hetzner	infra-hetzner/ansible/ — Ansible playbook	ansible-playbook site.yml from a clean control host

Table 2 — Reproducible infrastructure entry points for AWS and Hetzner.

Both halves are run from an operator's machine — no manual one-off configuration steps are required. The CDK app picks up environment specifics from `infra/envs/production.json`; the Ansible playbook is self-contained.

Key in-repo entry points:

- [infra/](#) — CDK app, stacks under `infra/src/lib/stacks/`
- [infra-hetzner/ansible/site.yml](#) — Hetzner provisioning entry point

AC 5 — Monitoring and alerting

"CloudWatch dashboard accessible; Galexie lag alarm fires correctly in staging."

Dashboard: `production-soroban-explorer` (AWS CloudWatch, eu-central-1). The ingestion section shows Galexie freshness (time since last ledger exported) and the Ledger Processor's invocation duration and error rate on a single pane.

Alarms (both green at the time of submission):

Alarm	Trigger
production-galexie-ingestion-lag	Fires if no new ledger is exported within the lag threshold.
production-indexer-ch-write-failures	Fires if the Lambda fails to write to ClickHouse.

Table 3 — Production monitoring alarms relevant to Milestone 1 ingestion and ClickHouse writes.

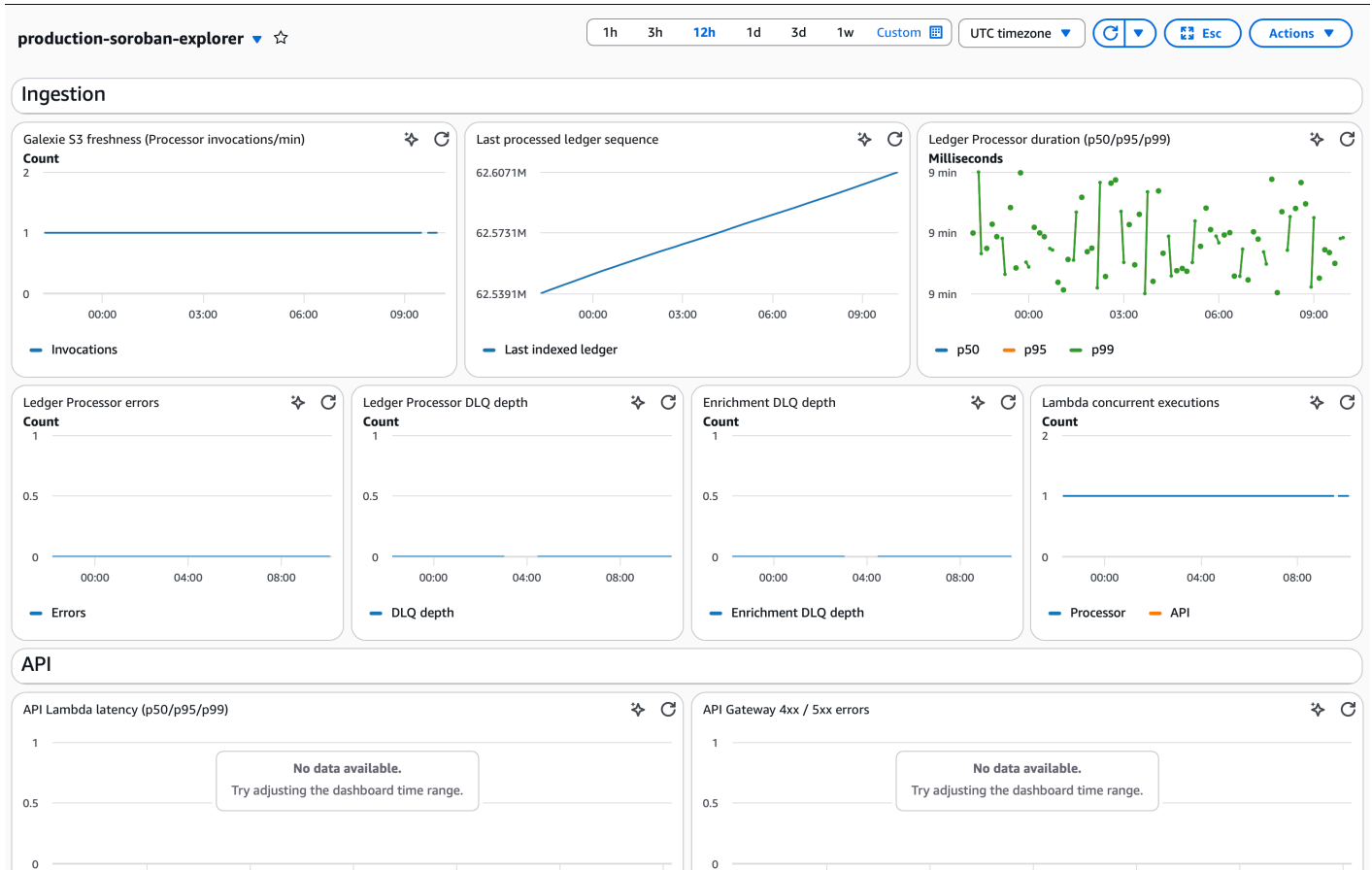


Figure 5 — CloudWatch dashboard *production-soroban-explorer* shows ingestion and API health signals over a 12-hour window.

CloudWatch > Alarms

Alarms (7) CloudWatch console feature Clear selection Create composite alarm Actions Create alarm

Filter alarms Quick filters

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	production-galaxie-ingestion-lag	OK	Actions enabled	-	2026-05-28 09:55:48	Invocations < 1 for 1 datapoints within 15 minutes
☐	production-ledger-processor-dlq-depth	OK	Actions enabled	-	2026-05-27 11:48:56	DLQ depth > 0 for 1 datapoints within 1 minute
☐	production-ledger-processor-error-rate	OK	Actions enabled	-	2026-05-27 11:45:38	Error Rate > 0.01 for 1 datapoints within 5 minutes
☐	production-indexer-ch-write-failures	OK	Actions enabled	-	2026-05-27 11:41:20	ChWriteFailures > 10 for 1 datapoints within 5 minutes
☐	production-api-gateway-5xx-rate	OK	Actions enabled	-	2026-05-27 09:25:32	5xx Rate (%) > 0.5 for 1 datapoints within 5 minutes
☐	production-enrichment-worker-error-rate	OK	Actions enabled	-	2026-05-26 11:29:07	Worker Error Rate > 0.01 for 1 datapoints within 5 minutes
☐	production-enrichment-dlq-depth	OK	Actions enabled	-	2026-05-26 11:27:55	Enrichment DLQ depth > 0 for 1 datapoints within 1 minute

Figure 6 — CloudWatch alarms list shows the production ingestion, ClickHouse write, API, and enrichment alarms in OK state.

6. Live endpoints and access

Resource	URL / address	Access
Production ClickHouse	ch.sorobanscan.rumblefish.dev	mTLS — client certificate on request
Production CloudWatch dashboard	production-soroban-explorer (eu-central-1)	IAM, read-only access on request
GitHub repository (public)	https://github.com/rumblefishdev/soroban-block-explorer	Anonymous
Project task ledger (lore board)	https://rumblefishdev.github.io/soroban-block-explorer/	Anonymous

Table 4 — Live verification endpoints and access model for reviewers.

Reviewers who want hands-on access to the private endpoints (production ClickHouse, production CloudWatch) can request a short-lived mTLS client certificate or a read-only IAM principal by emailing the address on the SCF submission form.

7. Repository navigation

For direct review of the implementation and design decisions, the relevant repository entry points are:

Topic	Path
Top-level architecture	docs/architecture/technical-design-general-overview.md
Database schema (DDL + field reference)	docs/architecture/database-schema/database-schema-overview.md
Indexing pipeline	docs/architecture/indexing-pipeline/
XDR parsing	docs/architecture/xdr-parsing/
Rust XDR parser library	crates/xdr-parser/
Ledger Processor Lambda	crates/indexer/
AWS CDK app	infra/
Hetzner Ansible playbook	infra-hetzner/ansible/site.yml
Lore: ADRs	lore/2-adrs/
Lore: completed M1 task ledger	lore/1-tasks/archive/ (tasks 0010–0042 and follow-ups)

Table 5 — Repository paths for implementation, infrastructure, architecture, and ADR review.

Key ADRs to read for Milestone 1 context:

- [ADR 0004](#) — Rust-only XDR parsing
- [ADR 0044](#) — ClickHouse pilot
- [ADR 0045](#) — backfill transport
- [ADR 0047](#) — ClickHouse as primary datastore